# MODULE 2: Data Link Layer
# (QB BY ME)

## Q1 What is link layer?

The **Data Link Layer** is the **2nd layer** in the **OSI model**, responsible for transferring datagrams across individual links between adjacent nodes. It defines packet formats and manages **error detection, retransmission, flow control, and random access**. Common protocols include **Ethernet, Token Ring, FDDI, and PPP**. A key feature is that different links in a path can use different link-layer protocols—for example, **Ethernet on one link and PPP on another**

### Q2 Services are provided by the Data Link Layer?

- **Framing:** The Data Link Layer encapsulates network layer data into **frames** before transmission, defining structure and access protocols for efficient delivery.
- **Reliable Delivery:** Ensures **error-free transmission** using **acknowledgments and retransmissions**, preventing unnecessary end-to-end retransmissions.
- **Flow Control:** Prevents sender overload by regulating the **data transmission rate**, avoiding receiver buffer overflow.
- **Error Detection & Correction:** Detects errors using **error detection bits** and, in some cases, **corrects** them without retransmission.
- **Half-Duplex & Full-Duplex:**

- **Full-Duplex:** Both nodes can **transmit simultaneously**.
- **Half-Duplex:** Only one node can **transmit at a time**.
- **Error Control:** Ensures frames are transmitted with **accuracy** by detecting and correcting errors.

### Q3 Errors in Data Transmission

When data is transmitted over a network, **interference and network issues** can corrupt bits, leading to errors. These errors result in **incorrect data** being received at the destination.

**Types of Errors:**

1. **Single Bit Error** – Only **one bit** in the frame is altered (0 → 1 or 1 → 0).
2. **Multiple Bit Error** – More than **one non-consecutive** bits are corrupted.
3. **Burst Error** – **Multiple consecutive** bits are corrupted, making it more severe.

### Q4. What is Error Control and Redundancy?

**Redundancy & Error Detection**

- **Redundancy** involves adding extra (redundant) bits to detect errors in data transmission.
- These bits help the receiver identify errors and remove them if possible.

**Error Control**

- **Detection vs. Correction:**
  - **Detection** checks if errors exist (Yes/No).
  - **Correction** requires identifying and fixing the corrupted bits.
- **Forward Error Correction (FEC) vs. Retransmission:**
  - **FEC**: The receiver **guesses** missing/corrupted data using redundant bits, without asking the sender to resend.
  - **Retransmission (Backward Error Correction)**: If an error is found, the receiver **requests** the sender to resend the entire message.

**Q5. What is Error-Detection and Correction Techniques?**

Error detection and correction techniques help ensure accurate data transmission by identifying and fixing errors. These methods use **redundancy bits** to verify data integrity.

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## Error Detection Techniques:

1. **Simple Parity Check:**
   - Adds an extra bit to maintain even parity (even number of 1s).
   - **Drawback:** Can only detect single-bit errors, not multiple-bit errors.
2. **Two-Dimensional Parity Check:**
   - Uses parity bits for both rows and columns in a data block.
   - Enhances error detection capability.
3. **Checksum:**
   - Divides data into segments, computes a sum using **1's complement**, and sends it as a checksum.
   - At the receiver's end, if the recalculated sum is **zero**, data is correct.
4. **Cyclic Redundancy Check (CRC):**
   - Uses **binary division** instead of addition.
   - Sender appends CRC bits to make the data **exactly divisible** by a predefined binary number.
   - Receiver checks for remainder; **zero remainder** means no errors.

## Error Correction Techniques:

5. **Hamming Code:**
   - Uses **parity bits at specific positions** to detect and correct single-bit errors.
6. **Forward Error Correction (FEC):**
   - Sends redundant data so the receiver can correct errors without retransmission.
7. **Automatic Repeat Request (ARQ):**
   - Requests retransmission if errors are detected using techniques like **Stop-and-Wait ARQ and Go-Back-N ARQ**.

**Q6. What is Framing in Data Link Layer?**

**Framing** is the process of dividing a continuous stream of bits into **discrete, manageable blocks** called **frames** for transmission in a network. Frames are units of digital transmission that organize data for efficient communication. Frames are similar to photons in light energy—they carry data in **structured packets.** Each frame contains **header, payload (data), and trailer**. The **header** includes **source and destination addresses** and error-checking codes. Frames help detect and correct errors using **Cyclic Redundancy Check (CRC)** or **checksums**. Ensures **organized data transfer** between two devices.

## 1. Fixed-Size Framing

- Each frame has a **fixed length**, so no explicit boundaries are needed.
- **Drawback:** Internal fragmentation if data is smaller than the frame size.
- **Solution: Padding** is used to fill the unused space.

## 2. Variable-Size Framing

- Frames can have **different lengths**, so clear boundaries are required.
- Two methods to define frame boundaries:

**a) Length Field:**

- Specifies the frame's length (**used in Ethernet 802.3**).
- **Issue:** If corrupted, frame boundaries become unclear.

**b) End Delimiter (ED):**

- Uses a **special pattern** to mark frame boundaries (**used in Token Ring**).
- **Issue:** If ED appears in data, it can cause errors.
- **Solutions:**
    - **Character Stuffing:** Adds an escape character (\○) before ED in character-based frames.
    - **Bit Stuffing:** Inserts a **0** into the data if it matches ED, ensuring proper detection

## Q7 Noiseless and Noisy Channels (5 Marks)

Data link layer protocols are categorized based on the presence of **errors** in the transmission channel:

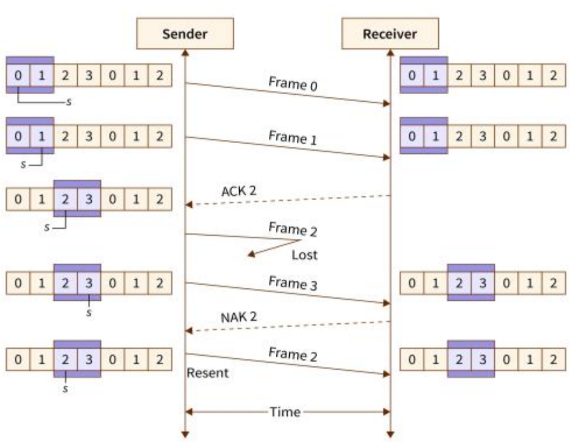
### 1. Noiseless Channels

- **Assumption:** The transmission is **error-free** (ideal conditions).
- **Protocol Used: Stop-and-Wait Protocol**
  **Working:**
    - Sender transmits a **single frame** and waits for an acknowledgment (ACK).
    - Receiver processes the frame and sends **ACK**.
    - Sender sends the next frame **only after** receiving the ACK.
    - Process repeats until the **End of Transmission (EoT)** frame is sent.
    - **Limitation:** Inefficient for high-speed networks due to waiting time.

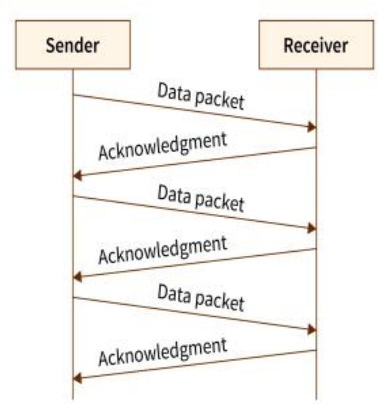
### 2. Noisy Channels

- **Assumption:** Errors occur due to **interference, noise, or signal distortion**.
- **Protocol Used: Sliding Window Protocol**
  **Working:**
    - The sender maintains a **window** that allows sending **multiple frames** before receiving ACKs.
    - The receiver has a **fixed window size (usually 1 frame)**.
    - Each frame is numbered **(0 to n-1)**, where **n** is the sender's window size.
    - After receiving a certain number of frames, the receiver sends an **ACK** with the next expected frame number.
    - **Advantage:** Increases efficiency by allowing continuous transmission.

These protocols ensure **smooth data transmission** based on channel conditions.




STOPN-AND-WAIT PROTOCOL



**Data Rate (R) = Total number of bits transmitted / Total time taken to transmit the bits**

## Difference Between Flow Control and Error Control

| Feature | Flow Control | Error Control |
|---|---|---|
| **Definition** | Manages the transmission rate between sender and receiver. | Ensures error-free and reliable data transmission. |
| **Purpose** | Prevents data loss due to receiver overload. | Detects and corrects errors in data frames. |
| **Approaches Used** | - **Feedback-based Flow Control** - **Rate-based Flow Control** | - **Error Detection:** Checksum, CRC, Parity Checking - **Error Correction:** Hamming Code, Reed-Solomon, LDPC |
| **Focus Area** | Maintains **proper data flow** between sender and receiver. | Ensures **error-free** data transmission. |
| **Examples** | - Stop-and-Wait Protocol - Sliding Window Protocol | - Stop-and-Wait ARQ - Sliding Window ARQ |

## Q8 High-Level Data Link Control (HDLC) (5 Marks)

HDLC is a **bit-oriented** data link layer protocol used for **reliable data transmission** between network nodes.

✔ **Frame-Based Transmission:** Data is organized into **frames** for error detection and reliable delivery.

✔ **Supports both Point-to-Point & Multipoint Communication.**

✔ **Error Detection & Flow Control:** Ensures data integrity.

| Mode | Description |
|---|---|
| **Normal Response Mode (NRM)** | - **Master-Slave** system: A **primary** station sends commands, and a **secondary** station responds. - Used in **point-to-point** and **multipoint** communication. |
| **Asynchronous Balanced Mode (ABM)** | - **Peer-to-Peer** system: Both stations can send and receive commands. - Used **only in point-to-point** communication. |

The fields of a HDLC frame are –

➢ Flag – It is an 8-bit sequence that marks the beginning and the end of the frame.

➢ Address – It contains the address of the receiver. If it is sent by the secondary station, it contains the address of the primary station. The address field may be from 1 byte to several bytes.

➢ Control – It is 1 or 2 bytes containing flow and error control information.

➢ Payload – This carries the data from the network layer. Its length may vary from one network to another.

➢ FCS – It is a 2 byte or 4 bytes frame check sequence for error detection

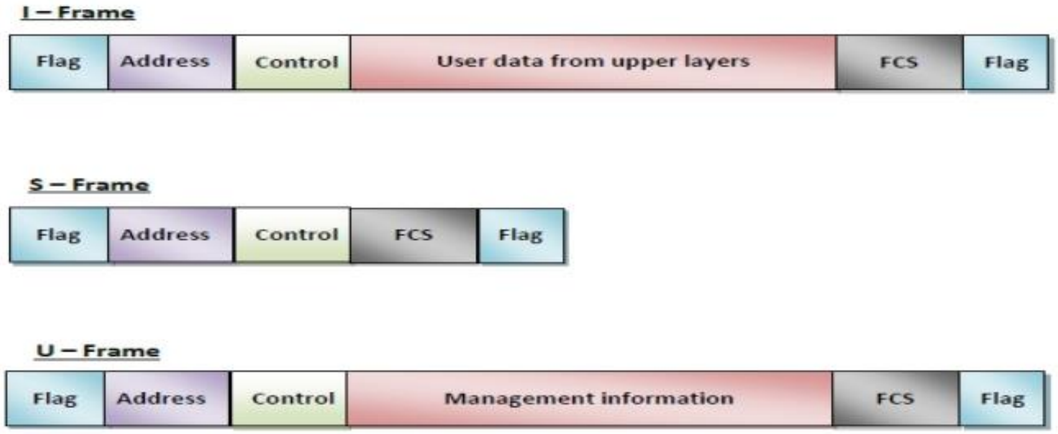
## Q9 What are Types of HDLC Frames ?

There are three types of HDLC frames. The type of frame is determined by the control field of the frame.

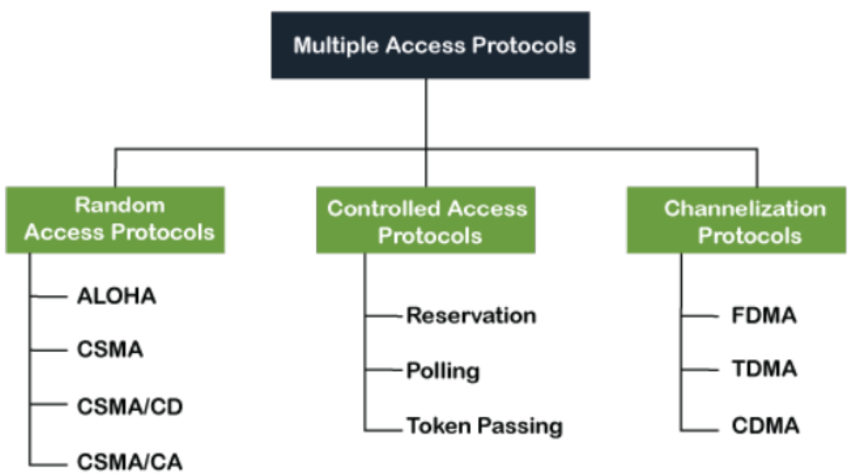
➢ I-frame − I-frames or Information frames carry user data from the network layer. They also include flow and error control information that is piggybacked on user data. The first bit of control field of I-frame is 0.

➢ S-frame − S-frames or Supervisory frames do not contain information field. They are used for flow and error control when piggybacking is not required. The first two bits of control field of S-frame is 10.

➢ U-frame − U-frames or Un-numbered frames are used for myriad miscellaneous functions, like link management. It may contain an information field, if required. The first two bits of control field of U-frame is 11




## Q10 Multiple Access Protocols? (10M) (IMP)




### 1. Random Access Protocols

- No fixed control over transmission; stations transmit **whenever they want.**
- Collisions are possible, and mechanisms exist to **handle retransmissions.**

**Types of Random Access Protocols:**

✔ **ALOHA:**

- Any station can send data at any time.
- **Pure ALOHA:** Transmit & wait for an acknowledgment; retransmit after a random delay if no ACK.
- **Slotted ALOHA:** Time is divided into slots, and transmissions occur **only at the start of a slot**, reducing collisions.

✔ **CSMA (Carrier Sense Multiple Access):**

- Stations sense the channel before transmitting.
- **1-persistent:** Continuously senses the medium and transmits immediately when idle.
- **Non-persistent:** Waits randomly before rechecking if busy.
- **p-persistent:** Uses probability-based transmission in slots.

✔ **CSMA/CD (Collision Detection):**

- Used in **Ethernet.**
- Detects collision and stops transmission immediately, then retransmits after a random delay.

✔ **CSMA/CA (Collision Avoidance):**

- Used in **Wi-Fi.**
- Avoids collision by waiting before transmission (Interframe Space).

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**2. Controlled Access Protocols**

- Stations take **permission** before sending data.
- No collisions occur.

**Types of Controlled Access Protocols:**

✔ **Reservation:** Stations reserve slots in advance.

✔ **Polling:** A central controller asks each station if they want to send data.

✔ **Token Passing:** A **token (control signal)** is passed around, and only the station holding the token can transmit.

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**3. Channelization Protocols**

- **Bandwidth is divided** among users based on frequency, time, or coding techniques.

**Types of Channelization Protocols:**

✔ **FDMA (Frequency Division Multiple Access):**

- **Each user gets a separate frequency band.**
- Used in **radio & TV broadcasting.**

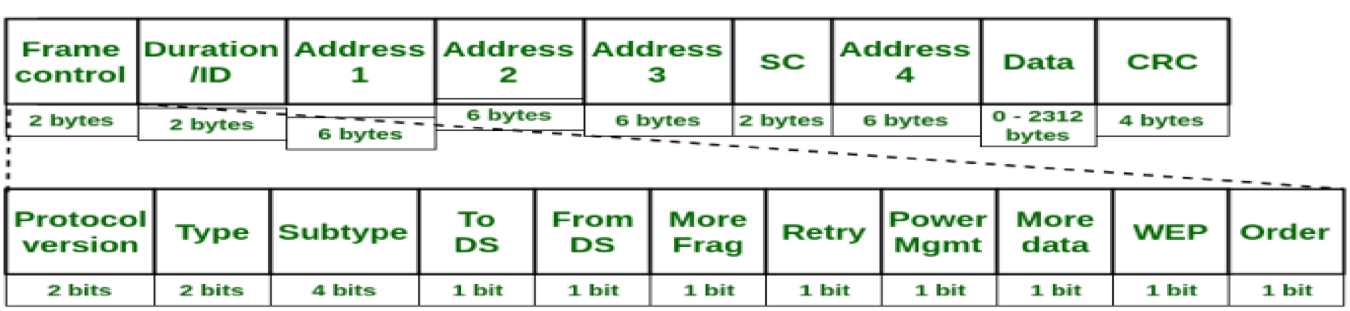
✔ **TDMA (Time Division Multiple Access):**

- **Same frequency shared** by dividing time into slots.
- Used in **2G & 3G cellular networks.**

✔ **CDMA (Code Division Multiple Access):**

- **All users share the same frequency & time,** but each data stream uses a unique code.
- Used in **4G & 5G networks.**

Q11




IEEE 802.11 MAC Frame Structure

1. **Introduction**
    - The IEEE 802.11 MAC frame is the **basic unit of data transfer** in Wi-Fi networks.
    - It contains **control information**, **addresses**, **data**, and **error-checking** fields.
2. **Frame Control (2 bytes)**
    - Determines **frame type** (management, control, or data) and includes protocol version.
    - Houses various control bits such as **Retry**, **Power Management**, **WEP**, etc.
3. **Duration/ID (2 bytes)**
    - Specifies how long the **wireless medium** will be occupied for this frame (in microseconds).
    - Sometimes used as an ID field for special control frames.
4. **Address Fields (Address 1 to 4, 6 bytes each)**
    - Up to **four MAC addresses** are used, depending on whether frames are going **to or from** the Distribution System (DS).
    - Typically includes **Source**, **Destination**, and possibly **Access Point** (BSSID) addresses.
5. **Sequence Control (2 bytes)**
    - Contains a **Sequence Number** (12 bits) to detect duplicates.
    - Contains a **Fragment Number** (4 bits) for reassembling fragmented frames.
6. **Data (Frame Body)**
    - Variable-length field (up to ~2300 bytes).
    - Carries the **actual payload** (e.g., IP packets, TCP segments, or other user data).
7. **CRC (4 bytes)**
    - A **Cyclic Redundancy Check** that helps **detect errors** in the transmitted frame.
    - If the CRC fails at the receiver, the frame is discarded.
8. **Management Frames**
    - Used for **network setup** and maintenance (e.g., beacons, authentication, association).
    - Ensure devices can **discover** and **join** wireless networks.
9. **Control Frames**
    - Coordinate channel access and frame delivery (e.g., **RTS/CTS**, ACK).
    - Help **manage collisions** and ensure reliable transmission.
10. **Data Frames**

- Carry **user data** (actual information) between stations.
- This is the main payload in regular communication once devices are associated.

11. **Fragmentation**

- Large data packets can be **split (fragmented)** into multiple frames.
- Improves reliability by sending smaller chunks, reducing the impact of errors.

12. **Acknowledgments**

- After receiving a frame correctly, the receiver sends an **ACK**.
- If no ACK arrives, the sender **retransmits** the frame, ensuring reliability.